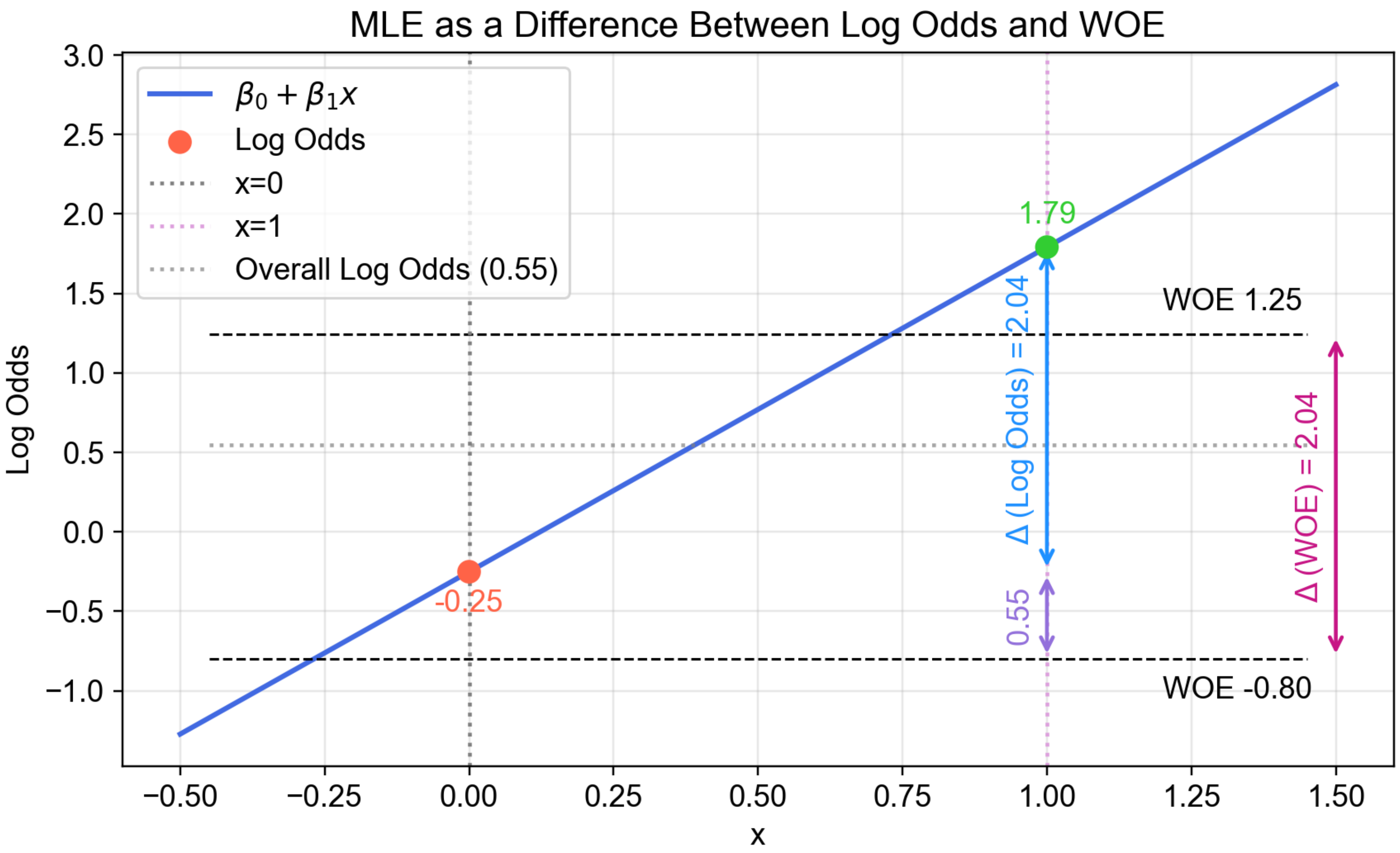
From Weight of Evidence to Logistic Regression

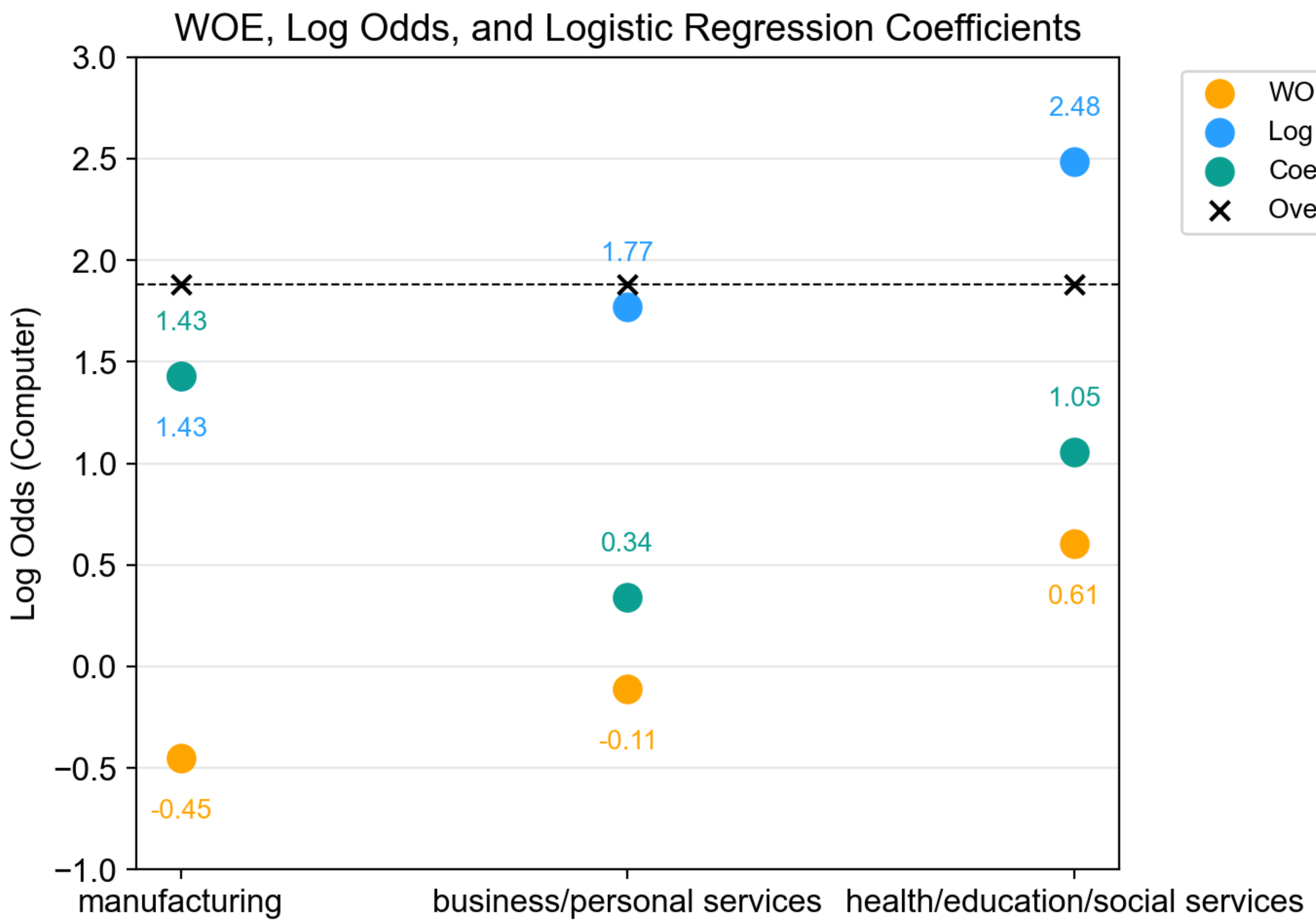
Did you know that there is a direct way to convert Weight of Evidence (WOE) scores to logistic regression estimates for univariate categorical data?

Univariate logistic regression (one category)*



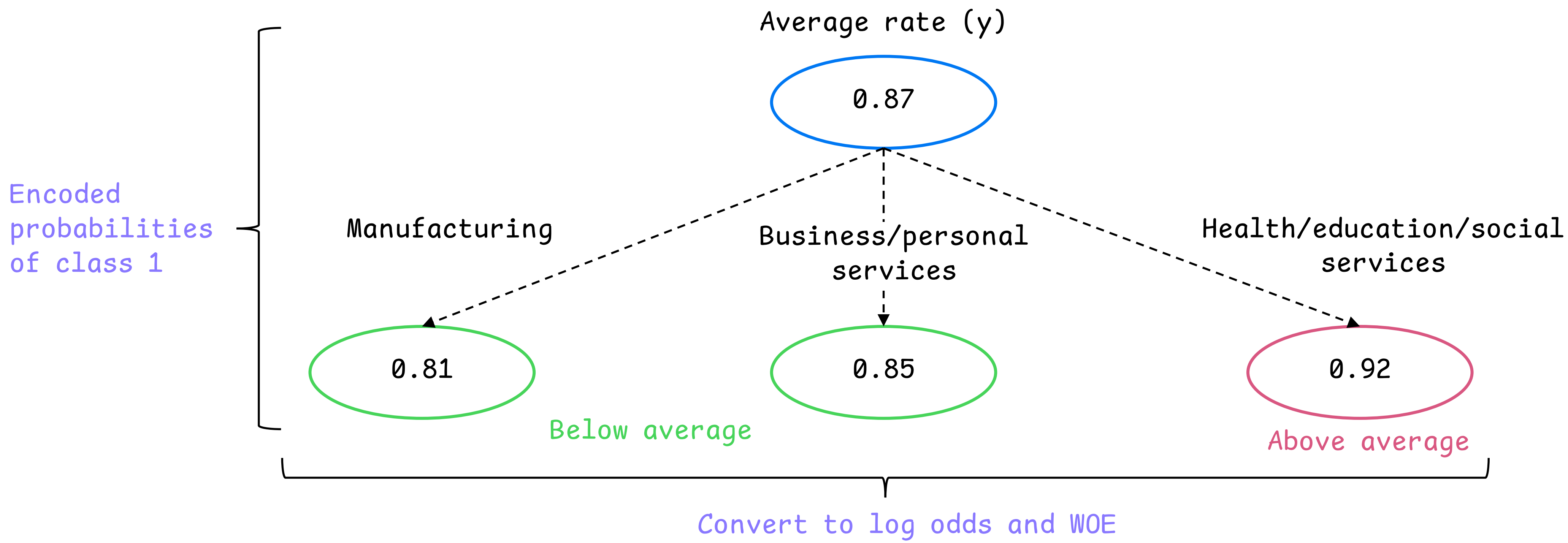
*See Appendix for a complete decomposition. Dataset is from https://academic.oup.com/aje/article-abstract/179/2/252/123902.

Univariate logistic regression (several categories)*



* Dataset used is ResumeNames from Applied Econometrics with R (AER). Target is `computer'.

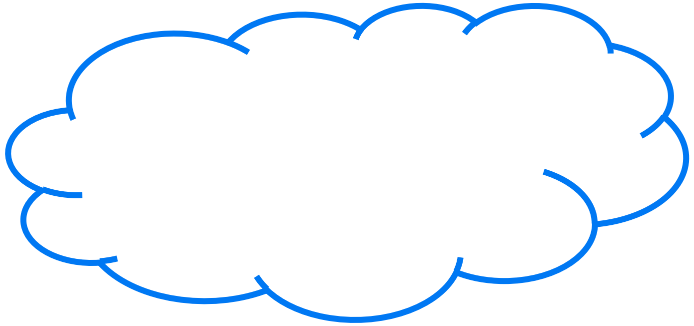
From univarite target encoding to logistic regression



Log odds	1.43	Δ=0.34	1.77	Δ=1.05	2.48
Weight of Evidence*	-0.45	Δ=0.34	-0.11	Δ=1.05	0.61

Logistic regression (MLE) coefficients: b0 (intercept) = 1.43, b1 = 0.34, b2 = 1.05 ✅

* WOE can be derived from two probabilities as log(0.81/(1-0.81)) - log(0.87/(1-0.87)) = -0.45



WOE is effectively another way to frame the evidence captured by logistic regression coefficients in the single feature case

Appendix

Notebook:
https://nbviewer.org/github/deburky/calibration/blob/main/logistic-regression-inference/from_woe_to_logistic_regression.ipynb

Univariate logistic regression (one category)

Has delinquency

Conditional probabilities (on x)

$P(Y|X)$

Data (Count)	y=1	y=0	Total
x=1	12	2	14
x=0	7	9	16
Total	19	11	30

Conditional probabilities (on y)

$P(X|Y)$

Data (Count)	y=1	y=0	Total
x=1	12	2	14
x=0	7	9	16
Total	19	11	30

Univariate logistic regression (one category)

Has delinquency

$P(Y|X)$

Conditional probabilities (on x)

	y=1	y=0	Total
x=1	86%	14%	100%
x=0	44%	56%	100%

$P(X|Y)$

Conditional probabilities (on y)

	y=1	y=0	Total
x=1	63%	18%	
x=0	37%	82%	
Total	100%	100%	

Univariate logistic regression (one category)

Has delinquency

Conditional probabilities (on x)

$P(Y|X)$

	y=1	y=0	Log odds
x=1	log(86%	/ 14%)	1.79
x=0	log(44%	/ 56%)	-0.25

Δ

Logistic Regression

Intercept	-0.25
b1	2.04 Δ

Difference b/w log odds or WOE for x=1 and x=0

Conditional probabilities (on y)

$P(X|Y)$

	y=1	y=0	WOE
x=1	log(63%	/ 18%)	1.25
x=0	log(37%	/ 82%)	-0.80

Δ